

KISHORE S

AI Engineer | Software Engineer

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[linkedin.com/in/kishoreselvaraj07](https://www.linkedin.com/in/kishoreselvaraj07) | github.com/kishorevijay07 | leetcode.com/u/kishorevijay978 (400+ solved)

PROFESSIONAL SUMMARY

AI and Software Engineer who ships production systems end to end — distributed backend services in Python and FastAPI, plus applied LLM systems covering RAG, multi-agent orchestration, and vector memory. Strong CS fundamentals (400+ DSA problems solved), with hands-on experience in REST API design, PostgreSQL tuning, React front ends, automated testing, and Dockerized CI/CD deployment.

EDUCATION

College of Engineering Guindy, Anna University — Chennai, India

Oct 2022 – May 2027

M.Sc. (Integrated) Information Technology

Relevant Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP, Machine Learning

EXPERIENCE

Software Developer Intern — IITM-Incubated Startup, Chennai

Jun 2025 – Nov 2025

- Developed and maintained backend booking management services in **Python and FastAPI**, implementing booking creation, status transitions, and fault-tolerant retry mechanisms.
- Designed **RESTful APIs** with input validation, idempotency keys, authentication, and structured error handling to ensure reliable and secure service-to-service communication.
- Optimized database performance by eliminating **N+1 query patterns** and applying indexing strategies, query tuning, and efficient pagination, improving application scalability and response times.
- Collaborated with frontend developers to integrate **React** applications with backend services, following **Agile/Scrum**, Git-based workflows, peer code reviews, and production deployment practices.

PROJECTS

TherapyOS — Multi-Tenant Practice-Management SaaS | FastAPI, PostgreSQL, Redis, Celery, React, Docker

- Founded and shipped a **multi-tenant SaaS platform** for pediatric therapy clinics, enforcing tenant isolation with PostgreSQL **Row-Level Security** across 16 tables and 3 role-based portals (Admin, Therapist, Parent); drove it from architecture to live clinic sales.
- Built **AI-assisted clinical documentation** generating SOAP notes and plain-language parent summaries in a single cached LLM call, cutting note-writing time by **~70%** while holding AI cost near **3% of revenue**.
- Engineered **WebSocket chat**, **Redis caching**, and payment webhooks and integrated **Razorpay billing and WhatsApp** messaging in a **Dockerized** deployment.

Another Me — Production-Grade Personal AI Assistant | Python, FastAPI, PostgreSQL/pgvector, Async SQLAlchemy, React, Docker, GitHub

- Architected and developed a full-stack personal AI assistant using **FastAPI, React, and TypeScript**, integrating LLM orchestration, **RAG**, semantic memory, and real-time response streaming.
- Engineered a **PostgreSQL/pgvector memory system** with relevance, importance, confidence, and recency-based retrieval, achieving **sub-second p95 latency** under concurrent load testing.
- Implemented an **intelligent LLM routing system** across four model tiers with automatic fallback, token-usage tracking, cost monitoring, and per-user budget enforcement via the **OpenRouter API**.
- Built production-grade infrastructure with **SSE streaming**, **JWT authentication**, **Redis rate limiting**, durable background job queues, Dockerized deployment.

AgentOS — Multi-Agent AI Operating System | Python, LangGraph, LangChain, FAISS, MCP, FastAPI

- Engineered an autonomous multi-agent system that plans, executes, and self-reflects on tasks, orchestrating **5 specialized agents** (Planner, Research, Coding, Testing, Documentation) as a **LangGraph** state machine.
- Integrated **OpenAI-compatible LLMs** with the **Model Context Protocol (MCP)** for standardized tool access, reducing manual input errors by **50%** and improving workflow efficiency on complex, multi-step inquiries.
- Implemented short- and long-term memory using **FAISS vector embeddings**, improving contextual continuity, reducing redundant retrieval, and enabling more efficient multi-agent task execution.

HR AssistHub — HR Automation Platform | React, FastAPI, LLM, RAG (Team of 4)

- Delivered an end-to-end platform automating employee communication and meeting coordination, replacing a manual workflow of **6 hours/week**.
- Integrated **LLM + RAG** over internal HR documents for automated email drafting and natural-language meeting scheduling, generating context-aware drafts in under **5 seconds**.

TECHNICAL SKILLS

Languages: Python, JavaScript, C++, SQL

AI & Machine Learning: LLMs, RAG, LangChain, LangGraph, Model Context Protocol (MCP), Prompt Engineering, Vector Databases (pgvector, FAISS), BERT Fine-Tuning, LoRA/QLoRA, CNNs, LSTMs, NER, Regression, Clustering

Backend, APIs & Frontend: FastAPI, Node.js, Express.js, RESTful API Design, WebSockets, Celery, Redis, Distributed Systems, Scalable System Design, React.js, Tailwind CSS

Databases: PostgreSQL, MySQL, MongoDB, pgvector, FAISS, Row-Level Security, Relational Schema Design, Query Optimization

DevOps, Tools & Practices: Docker, Git, GitHub Actions (CI/CD), MLflow, HuggingFace Transformers, Postman, Linux, SDLC, Unit & Integration Testing, Code Review

ACHIEVEMENTS

- Solved **400+ Data Structures & Algorithms problems** on LeetCode.
- Runner-Up, Xackthon Hackathon** (2nd of 40 teams); **Active Performer Award** for contribution and teamwork, CEG.